

FreshMarket – Enterprise Food Marketplace Platform

Project Summit Accelerated Candidate Program (Foundation Track)

- Candidate Name: Lochini Attanayake
- Date Started: July 2026
- Repository: <https://github.com/lochiniattanayake496-stack/Food-Marketplace-Platform>

Project Status & Milestones

Phase 1: Foundation & Planning — *In Progress (Repo Created, Monorepo Setup, Progress Deck Initialized)*

Phase 2: Architecture & Contracts — *Upcoming*

Phase 3: Full-Stack Implementation — *Upcoming*

Phase 4: Quality & Cloud Integration — *Upcoming*

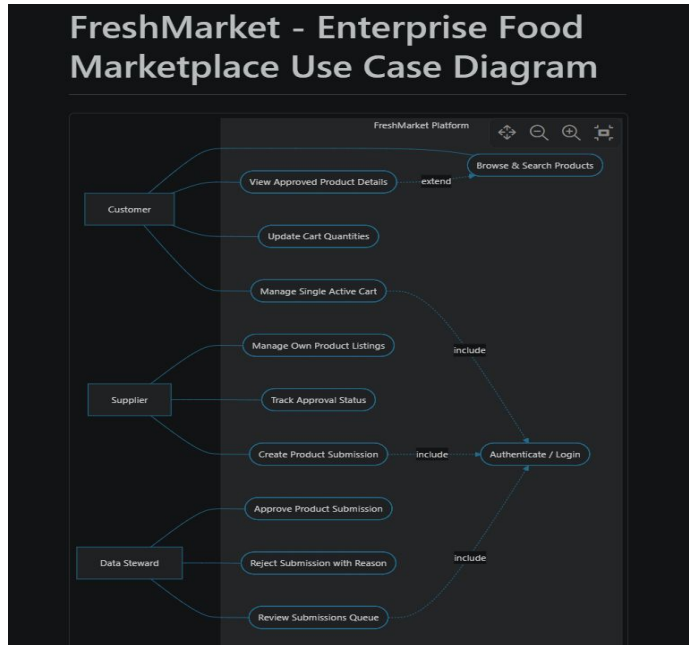
Phase 5: 4IR Innovation MVP — *Upcoming*

Technical Architecture & Choices

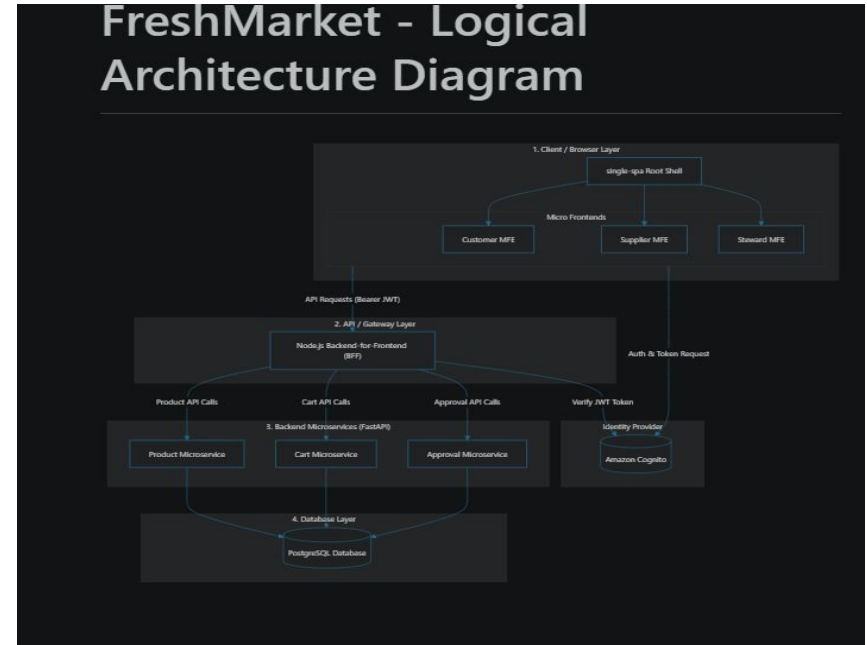
- **Frontend:** React.js + Redux + Micro Frontend Architecture ([single-spa](#))
- **BFF (Backend-for-Frontend):** Node.js (Express) for request aggregation and Cognito token validation
- **Backend Microservices:** Python 3.11 + FastAPI + SQLAlchemy ORM + PostgreSQL
- **Authentication:** Amazon Cognito (User Pool & Groups / RBAC)
- **Deployment & Cloud:** Dockerized Containers deployed to AWS

Architecture Diagrams & Data Flow

Use Case Diagram (User Personas, Actions, System Boundaries)

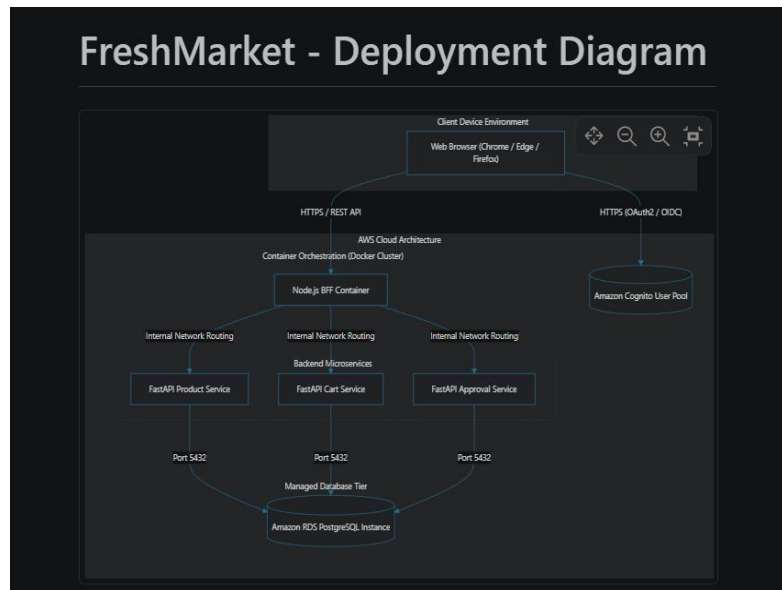


Logical Architecture Diagram (*single-spa - Node.js BFF - FastAPI Microservices -PostgreSQL*)



AWS Deployment & API Strategy

Deployment Diagram (AWS Cloud topology, Docker Containers on ECS, Amazon RDS for PostgreSQL, Cognito Auth).



Design-First API Strategy: Drafted OpenAPI 3.0 (.yaml) contracts for **Product**, **Cart**, and **Approval** services prior to backend implementation.

Core Endpoint Contracts:

- **product-service-api.yaml:** Catalog browsing, query filtering, supplier submission routes.
- **cart-service-api.yaml:** Single active cart state, quantity validation, item mutations.
- **approval-service-api.yaml:** Data Steward review queues, approval actions, rejection reason logging.

Practical Alignment: AWS infrastructure directly models actual containerized ECS services and managed RDS databases.